

江西海瑞天然植物有限公司
JIANGXI HAIRUI NATURAL PLANT CO. LTD.

ADD: No.267 JiaHua Avenue,Jinggangshan Economic Development Zone, Ji'an County,Ji'an City, Jiangxi
Zip code:343100 Tel: 0796-8507999 FAX: 0796-8108396 Website: www.jxhairui.com

MATERIAL SAFETY DATA SHEET OF

Polysorbate 80 (Tween 80)

1. CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

Product Name: Polysorbate 80 (Tween 80)
Manufacture: Jiangxi Hairui Natural Plant Co. Ltd.
Address: No.267 JiaHua Avenue,Jinggangshan Economic Development Zone,
Ji'an County,Ji'an City, Jiangxi , P.R.C
Post code: 343100
Fax: 86-0796-8507999
Tel: 86-0796-8107398

2. COMPOSITION/INGREDIENT INFORMATION

Ingredients: Polysorbate 80 (Tween 80)
Chemical Name: Polyoxyethylene (20) Sorbitan Monooleate
Product Type: Non-ionic Surfactant
Synonyms: Polyoxyethylene Sorbitan Monooleate
CAS #: 9005-65-6
EC Number: 500-019-9

3. HAZARDS IDENTIFICATION

3.1 Classification of Hazards

Skin Irritation: Category 3 (Mild Irritation)
Eye Irritation: Category 2B (Mild Irritation)
Specific Target Organ Toxicity (Single Exposure): Category 3 (Respiratory
Tract Irritation)

3.2 Label Elements

- **Signal Word:** WARNING
- **Pictogram:** Exclamation Mark (Irritant)
- **Hazard Statements**

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- H316: Causes mild skin irritation
- H320: Causes eye irritation
- H335: May cause respiratory irritation
- **Precautionary Statements**
- P261: Avoid breathing vapor/mist
- P264: Wash skin thoroughly after handling
- P280: Wear protective gloves/eye protection
- P302+P352: IF ON SKIN: Wash with plenty of soap and water
- P337+P313: If eye irritation persists: Get medical advice/attention
- P403+P233: Store in a well-ventilated place. Keep container tightly closed

3.3 Other Hazards

- No fire or explosion hazard; not classified as dangerous goods
- May oxidize during long-term storage to form peroxides

4. FIRST AID MEASURES

4.1 Skin Contact

Immediately remove contaminated clothing and rinse skin thoroughly with plenty of running water for at least 15 minutes. If irritation, dryness or rash occurs, apply moisturizer; seek medical attention if symptoms persist.

4.2 Eye Contact

Immediately hold eyelids open and rinse continuously with plenty of running water or normal saline for 15 minutes. Avoid rubbing eyes; seek medical attention if eye discomfort persists.

4.3 Inhalation

Quickly move the victim to fresh air and keep the respiratory tract unobstructed. If breathing is difficult, provide oxygen. If breathing stops, perform artificial respiration immediately and seek medical attention.

4.4 Ingestion

Do NOT induce vomiting. If the patient is conscious, rinse the mouth with warm water and give a small amount of milk or water to drink. Never give anything by mouth to an unconscious person. Seek medical attention if any discomfort occurs.

4.5 Most Important Symptoms and Effects

Skin contact may cause dryness and mild erythema; eye contact may cause redness and tearing; inhalation may cause coughing and chest tightness.

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5. FIRE FIGHTING MEASURES

5.1 Suitable Extinguishing Media

Water spray, dry chemical, carbon dioxide, foam (non-prohibited); **no special extinguishing agent is required** as this product is basically non-combustible.

5.2 Hazardous Combustion Products

Combustion produces carbon monoxide (CO), carbon dioxide (CO₂) and a small amount of irritating fumes.

5.3 Fire-Fighting Precautions and Protective Measures

- Firefighters must wear self-contained breathing apparatus (SCBA) and full fire-resistant protective clothing, and fight fire from the upwind direction.
- Cool containers in the fire scene with water spray until the fire is completely extinguished.

This product is **non-flammable**; the main hazard is fumes generated during combustion.

6. ACCIDENTAL RELEASE MEASURES (STEPS FOR SPILLS)

6.1 Personal Precautions

Eliminate all ignition sources and evacuate personnel to a safe area. Wear protective gloves and goggles; wear a dust mask if necessary.

6.2 Environmental Precautions

Prevent spilled material from entering sewers, rivers, soil and other environments. Small spills can be absorbed with dry sand or inert adsorbent materials.

6.3 Methods and Materials for Containment and Cleaning Up

- Small spills: Absorb with sand or other inert materials, collect in a sealed container for subsequent disposal.
- Large spills: Build dikes or dig pits to contain the spill, transfer to tank trucks or special collectors with a pump for recovery or disposal in accordance with regulations.
- Rinse the spill area with plenty of water after cleaning.

7. HANDLING AND STORAGE

7.1 Handling Precautions

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- Handle only in well-ventilated areas to avoid vapor accumulation.
- No smoking, eating or drinking during handling.
- Avoid direct contact with eyes and skin.
- Wash hands and exposed skin thoroughly after handling.

7.2 Storage Precautions

- Storage Temperature: 15-25°C, store in a sealed container away from light.
- Storage Container: Sealed HDPE plastic or glass container.
- Storage Area: Keep away from heat sources, open flames and oxidizing agents; store separately from strong acids and alkalis.
- Shelf Life: 24 months (under proper storage conditions).
- Prevent moisture from entering and avoid oxidative deterioration.

8. EXPOSURE CONTROL/PERSONAL PROTECTION

8.1 Occupational Exposure Limits

- No officially specified occupational exposure limits; refer to general management standards for non-ionic surfactants.

8.2 Engineering Controls

Adopt local exhaust ventilation or general ventilation to maintain air circulation in the workplace; regularly monitor vapor concentration in the working area.

8.3 Personal Protective Equipment (PPE)

- **Hand Protection:** Wear nitrile or neoprene gloves (avoid latex gloves).
- **Eye Protection:** Wear chemical splash goggles.
- **Skin Protection:** Wear impervious work clothes to avoid skin exposure.
- **Respiratory Protection:** Wear a dust mask or organic vapor filtering respirator if dust or vapor concentration exceeds the limit.

9. PHYSICAL AND CHEMICAL PROPERTIES

Item	Parameter
Appearance and State	Amber to yellow viscous liquid with slight characteristic odor
Flash Point (Closed Cup)	>110°C (basically non-combustible)

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Item	Parameter
Boiling Point	>300°C (decomposes)
Relative Density (Water=1)	1.06-1.09 (25°C)
Refractive Index (20°C)	1.471-1.475
Solubility	Easily soluble in water, ethanol and isopropanol; dispersible in mineral oil
Viscosity	300-600 mPa • s (25°C)
Freezing Point	-21°C

Composition/Information on Ingredients

Ingredient Name	CAS Number	Weight Percentage	Hazard Classification
Polyoxyethylene (20) Sorbitan Monooleate	9005-65-6	100%	H316, H320, H335
Total	—	100%	—

10. STABILITY AND REACTIVITY

10.1 Stability

Stable under normal storage and handling conditions.

10.2 Conditions to Avoid

High temperature (>60°C), direct sunlight, strong oxidizing agents, strong acids, strong alkalis.

10.3 Incompatible Materials

Strong oxidizing agents (e.g., hydrogen peroxide, potassium permanganate), strong acids, strong alkalis.

10.4 Hazardous Reactions

May react with strong oxidizing agents to cause product deterioration; no risk of violent reaction.

10.5 Hazardous Decomposition Products

May decompose to form peroxides under long-term storage or high-temperature conditions, which have mild irritation.

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11. TOXICOLOGICAL INFORMATION

11.1 Acute Toxicity

- Oral LD50 (Rat): >30,000 mg/kg (practically non-toxic)
- Dermal LD50 (Rabbit): >20,000 mg/kg (practically non-toxic)
- Inhalation LC50 (Rat, 4h): >5 mg/L (practically non-toxic)

11.2 Skin Corrosion/Irritation

Rabbit dermal test: Mild irritation, erythema subsides within 24 hours.

11.3 Skin Sensitization

Human patch test: Basically non-sensitizing; mild reactions may occur in a very small number of allergic individuals.

11.4 Other Toxicological Effects

- Subchronic Toxicity: 90-day feeding test in rats shows no obvious toxic effects.
- Mutagenicity: AMES test is negative; no mutagenic risk.
- Carcinogenicity: Classified as Group 3 by IARC (not classifiable as to its carcinogenicity to humans); no evidence of carcinogenicity.

12. ECOLOGICAL INFORMATION

12.1 Aquatic Toxicity

Fish LC50 (96h): >100 mg/L; Algae EC50 (72h): >100 mg/L; **low toxicity to aquatic organisms.**

12.2 Biodegradability

Readily biodegradable (degradation rate >60% in 28 days, complying with OECD 301 standard).

12.3 Other Ecological Effects

Avoid large-scale spills into water bodies to prevent short-term impacts on aquatic plants. The product is biodegradable with minimal long-term environmental effects.

13. DISPOSAL CONSIDERATIONS

13.1 Waste Characteristic

Non-hazardous waste (in accordance with GB 34330-2017 and EU classification).

13.2 Waste Disposal Methods

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- Small amounts of waste: Disposed of by qualified waste treatment institutions, or discharged into sewers after dilution in accordance with local regulations.
- Large amounts of waste: Recycle for reuse or incinerate by professional waste treatment plants.
- Packaging disposal: Thoroughly clean empty containers for recycling or disposal as general waste.
- Note: Do not dump or bury at will; comply with local environmental protection regulations.

14. TRANSPORT INFORMATION

14.1 UN Proper Shipping Name

Not dangerous goods

14.2 UN Number

None (this product is not classified as dangerous goods)

14.3 Transport Hazard Class

None

14.4 Packing Group

None

14.5 Transport Precautions

- Transport vehicles must be equipped with leak emergency response equipment.
- Prevent exposure to sunlight, rain and high temperature during transportation.
- Handle with care during loading and unloading to avoid container damage.
- **No special transport conditions are required** for air, sea or land transportation.

15. REGULATORY INFORMATION

- EU REACH Regulation: Listed in the EC Inventory, complying with CLP classification and labeling requirements.
- U.S. TSCA: Listed in the TSCA Chemical Substance Inventory.
- China *Catalogue of Hazardous Chemicals*: Not listed; managed as a general chemical.
- China GB 25554-2010: National Food Safety Standard - Food Additive Polyoxyethylene (20) Sorbitan Monooleate (Tween 80).
- Others: Export shall comply with relevant chemical management regulations of the destination country.

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16. ADDITIONAL INFORMATION

16.1 Preparation Instructions

This SDS is prepared based on existing data, for reference only in safe handling, transportation and storage, and does not constitute a product quality guarantee.

16.2 Disclaimer

The company shall not be liable for any losses caused by improper use, storage or disposal of this product. Users shall verify the applicability according to the actual application scenarios.

Term of Validity: 2024-12-31